Muhammad Haider Zaidi

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github.com/chefhaider Bavaria, Germany 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash
ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
Tools & Platforms: Docker, Kubernetes, AWS SageMaker, FastAPI, Git, GitLab CI/CD, Airflow
Competence: Deep Learning, MLOps, Computer Vision, Perception, Segmentation, Object Detection, Trajectory Prediction, Time Series Prediction, Data Science, ETL, Model Deployment

Experience

German National Metrology Institute (PTB) **March 2024 – Present**
Student Assistant Researcher *Berlin, DE*

- Designed and evaluated advanced deep learning models for metrological applications, enhancing model performance, accuracy, and training efficiency, exploring novel architectures and physics-informed hybrid loss functions.
- Collaborated effectively with PhD researchers, contributing to a publication at the FLOMEKO conference, demonstrating scientific background in deep learning.
- Developed interactive dashboards for comprehensive data visualization and robust model performance monitoring.

Folio3 **March 2022 – Nov 2023**
Machine Learning Engineer *California, US (remote)*

- Enhanced image segmentation accuracy to 94% and boosted processing speed by 5x using advanced AI algorithms; deployed solutions with Docker and FastAPI for efficient model serving.
- Engineered a real-time object detection system using TensorRT, reducing compute cost and inference time by 6x, and managed end-to-end MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Developed proof-of-concept solutions leveraging YOLO, few-shot learning, and CNN architectures for client demonstrations, translating research into tangible products.

Manda GmbH **March 2024 - June 2024**
Werkstudent Python Developer *Nürnberg, DE*

- Developed full-stack web applications providing AI-based services using Python (Flask) and engineered an AI chatbot application leveraging the OpenAI API and advanced LLM frameworks.

Freelance **July 2020 – March 2022**

Data Engineer *Remote*

- Constructed robust web scrapers and ETL pipelines for efficient acquisition and automated data ingestion/cleaning workflows, significantly improving data quality for ML modeling.

Projects

CAFA 6: Protein Function Prediction **2022**

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, utilizing AlphaFold for advanced feature extraction, and increasing model accuracy by 4% through novel custom feature engineering.

Razanlytics: Real-time Bitcoin Price Forecasting **2022**

Thesis Project

- Developed a real-time Bitcoin price forecasting system, leveraging advanced time series prediction models, and implemented robust data pipelines for high-frequency model updates.

Saaz: AI Self-Playing Chitrali Sitar **2023**

Signal Processing, Music Generation

- Collaborated to engineer the world's first AI-driven self-playing Chitrali Sitar, integrating advanced signal processing with music generation LLMs.
- Achieved significant real-world impact with the project featured on Coke Studio, showcasing innovative application of AI in creative fields.

Hugging Face **Open-Source**

Contributor

- Co-authored comprehensive technical documentation and implementation for Zero-shot Image Classification on the Hugging Face Hub, contributing to open-source knowledge.

Education

Friedrich Alexander Universität **Present**

Master of Science in Artificial Intelligence *Erlangen, DE*

- Focused on Deep Learning, Computer Vision, and Reinforcement Learning for advanced AI applications.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor
- Specialized in core Computer Science principles with a strong foundation in algorithms and data structures.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera**: TensorFlow Developer Certificate | **Databricks**: Lakehouse Fundamentals | **Datacamp**: Computer Vision in Python